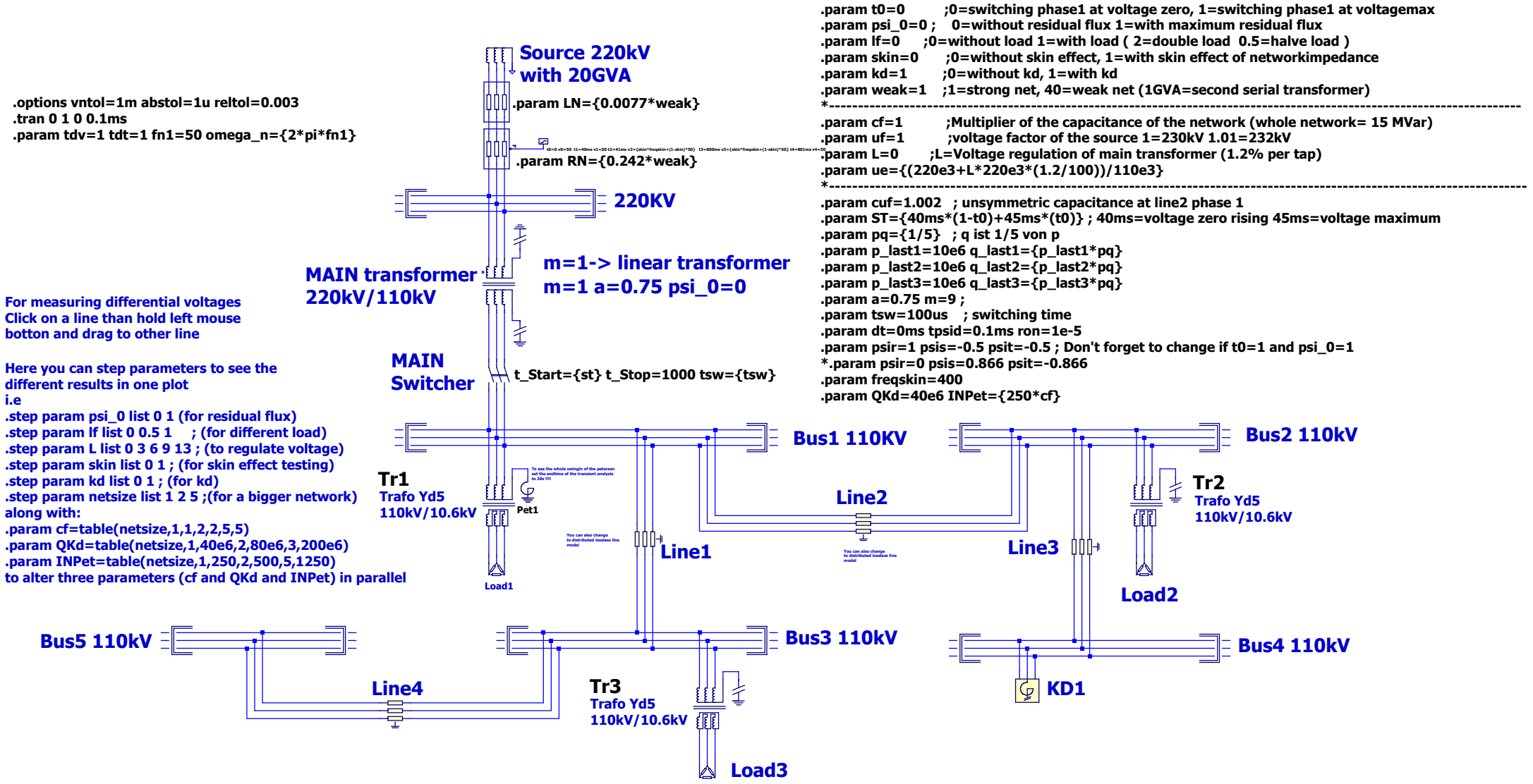
.options vntol=1m abstol=1u reltol=0.003
.tran 0 1 0 0.1ms
.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1}
Source 220kV
with 20GVA
.param LN={0.0077*weak}
.param RN={0.242*weak}
220KV
MAIN transformer
220kV/110kV
m=1-> linear transformer
m=1 a=0.75 psi_0=0
MAIN
Switcher
t_Start={st} t_Stop=1000 tsw={tsw}
Bus1 110KV
Bus2 110kV
Tr1
Trafo Yd5
110kV/10.6kV
Pet1
Line1
Line2
Line3
Tr2
Trafo Yd5
110kV/10.6kV
Load1
Load2
Bus5 110kV
Bus3 110kV
Bus4 110kV
Line4
Tr3
Trafo Yd5
110kV/10.6kV
KD1
Load3
.param t0=0 ;0=switching phase1 at voltage zero, 1=switching phase1 at voltagemax
.param psi_0=0 ; 0=without residual flux 1=with maximum residual flux
.param lf=0 ;0=without load 1=with load (2=double load 0.5=halve load)
.param skin=0 ;0=without skin effect, 1=with skin effect of networkimpedance
.param kd=1 ;0=without kd, 1=with kd
.param weak=1 ;1=strong net, 40=weak net (1GVA=second serial transformer)
.param cf=1 ;Multiplier of the capacitance of the network (whole network= 15 MVar)
.param uf=1 ;voltage factor of the source 1=230kV 1.01=232kV
.param L=0 ;L=Voltage regulation of main transformer (1.2% per tap)
.param ue={(220e3+L*220e3*(1.2/100))/110e3}
.param cuf=1.002 ; unsymmetric capacitance at line2 phase 1
.param ST={40ms*(1-t0)+45ms*(t0)} ; 40ms=voltage zero rising 45ms=voltage maximum
.param pq={1/5} ; q ist 1/5 von p
.param p_last1=10e6 q_last1={p_last1*pq}
.param p_last2=10e6 q_last2={p_last2*pq}
.param p_last3=10e6 q_last3={p_last3*pq}
.param a=0.75 m=9 ;
.param tsw=100us ; switching time
.param dt=0ms tpsid=0.1ms ron=1e-5
.param psir=1 psis=-0.5 psit=-0.5 ; Don't forget to change if t0=1 and psi_0=1
*.param psir=0 psis=0.866 psit=-0.866
.param freqskin=400
.param QKd=40e6 INPet={250*cf}
For measuring differential voltages
Click on a line than hold left mouse
botton and drag to other line
Here you can step parameters to see the
different results in one plot
i.e
.step param psi_0 list 0 1 (for residual flux)
.step param lf list 0 0.5 1 ; (for different load)
.step param L list 0 3 6 9 13 ; (to regulate voltage)
.step param skin list 0 1 ; (for skin effect testing)
.step param kd list 0 1 ; (for kd)
.step param netsize list 1 2 5 ;(for a bigger network)
along with:
.param cf=table(netsize,1,1,2,2,5,5)
.param QKd=table(netsize,1,40e6,2,80e6,3,200e6)
.param INPet=table(netsize,1,250,2,500,5,1250)
to alter three parameters (cf and QKd and INPet) in parallel